

# COMPRESSION COLUMN LOAD CELL FOR WEIGHBRIDGES/TRUCK SCALES

capacities 15t - 60t



This stainless steel self-centering rocker column load cell is fully welded and hermetically sealed, with protection to IP68 & IP69K. It's an ideal solution for weighbridges (truck scales) thanks to its approval to OIML C4 (4000 divisions) and integral lightning surge arrestors. As well as being extremely tough, the cable's halogen-free Polyurethane sheath has high resistance to rodents, chemical attack, cuts, ultraviolet light and temperature changes.

Corner adjustment at the junction box is minimised because the T34X has optimized pre-corner adjustment (calibrated in mV/V/ $\Omega$  by output current matching at  $\pm 0.05\%$ ). This saves a significant amount of time during commissioning.

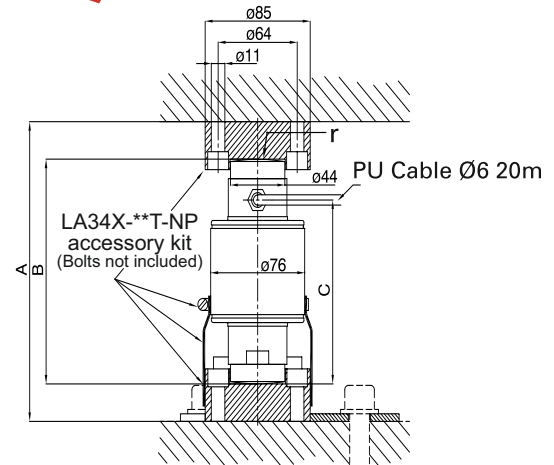
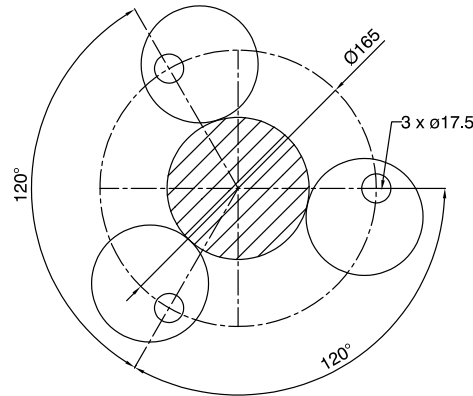
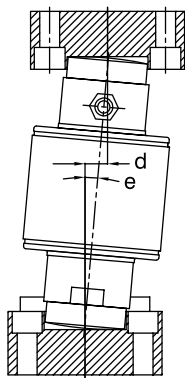
For use in weighbridges (truck scales) a mounting cup accessory set (LA34X) can be ordered separately. Made from nickel plated alloy steel, it consists of a rubber gaiter, a sealing clip and three centering discs.

For applications in aggressive environments where stress corrosion or acid attack could be a problem, a special Parylene coating can be specified as an option. Also available is an ATEX/IECEx version or a High Temperature version, allowing continuous use at up to  $+150^{\circ}\text{C}$ .

- Stainless steel load cell
- Fully welded and hermetically sealed to IP68 and IP69K
- In-built surge arrestors for lightning protection
- Industry standard height and mounting pattern
- Calibrated in mV/V/ $\Omega$ , by output current matching at  $\pm 0.05\%$ , to minimise corner adjustment
- Simple installation
- Self-centering rocker column design with anti-rotation faces
- Approval to OIML R60 C4 (4000 divisions)
- Wide operating temperature range
- ATEX/IECEx, High Temperature and Parylene coated options available
- 3 Year Warranty

# T34X

## technical specification...



Important: Bumper stops must be fitted to the weighbridge / truck scale and adjusted to limit movement to a recommended min. 4mm, max. 9mm in both the longitudinal and transverse directions. This is essential in order to avoid damage to the equipment. Refer to the critical dimensions d and e below (\*\*).

### T34X Load Cell

|                                                                                    | Load cell specification  | Units                |
|------------------------------------------------------------------------------------|--------------------------|----------------------|
| Load Cell Capacities (E <sub>max</sub> )                                           | 15, 20, 30, 40, 50, 60   | tonnes               |
| Rated Output (S <sub>n</sub> )                                                     | 2                        | mV/V*                |
| Accuracy Class                                                                     | 4000                     | n.OIML               |
| Combined Error                                                                     | < ± 0.013                | %S <sub>n</sub>      |
| Non-Repeatability                                                                  | < ± 0.01                 | %S <sub>n</sub>      |
| Creep (30 minutes)                                                                 | < ± 0.012                | %S <sub>n</sub>      |
| Minimum load cell verification interval (V <sub>min</sub> ) = E <sub>max</sub> / Y | E <sub>max</sub> / 10000 | kg                   |
| Temperature Effect on Zero Balance                                                 | < ± 0.002                | %S <sub>n</sub> / °C |
| Temperature Effect on Span                                                         | < ± 0.0012               | %S <sub>n</sub> / °C |
| Compensated Temperature Range                                                      | -10 to +40               | °C                   |
| Operating Temperature Range                                                        | -50 to +80               | °C                   |
| Safe Overload                                                                      | 150                      | % E <sub>max</sub>   |
| Ultimate Overload                                                                  | 350                      | % E <sub>max</sub>   |
| Zero Balance                                                                       | ≤ ± 2                    | %S <sub>n</sub>      |
| Input Resistance                                                                   | 800 ± 5                  | Ω                    |
| Output Resistance                                                                  | 705 ± 5                  | Ω                    |
| Insulation Resistance                                                              | > 5000                   | MΩ @ 100V            |
| Recommended Supply Voltage                                                         | 5-15                     | V                    |
| Maximum Supply Voltage                                                             | 15                       | V                    |
| Environmental Protection                                                           | IP68 and IP69K           |                      |
| Maximum deflection at E <sub>max</sub>                                             | 0.6-1                    | mm                   |
| Cable Length                                                                       | 20                       | m                    |
| Cable Type                                                                         | Polyurethane, 6 core     |                      |
| Nominal Shipping Weight                                                            | 15t, 20t                 | 2.5 kg               |
|                                                                                    | 30t                      | 2.6 kg               |
|                                                                                    | 40t                      | 2.8 kg               |
|                                                                                    | 50t                      | 2.9 kg               |
|                                                                                    | 60t                      | 3.8 kg               |
|                                                                                    | Mounting Accessories     | 3 kg                 |

\* Optimized pre-corner adjustment (Calibrated in mV/V/Ω by output current matching at ±0.05%)

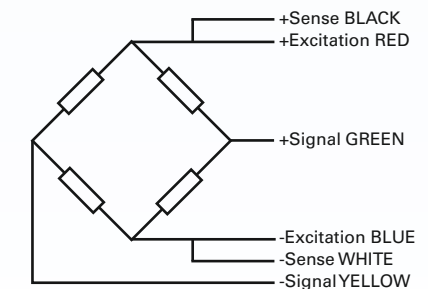
### Model T34X ATEX / IECEx Certification

| Code                                                                           | Specific parameters of protection types             | Application                                                                |
|--------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|
| II 1 G D<br>Ex ia IIC T4...T6 Ga<br>Ex ia IIIC T85°C Da<br>Ex ta IIIC T85°C Da | Protection type "Ex i"<br>Ex ia IIC T4 / Ex ia IIIC | Pi = 1.3W                                                                  |
|                                                                                | Protection type "Ex i"<br>Ex ia IIC T5              | Pi = 0.6W                                                                  |
|                                                                                | Protection type "Ex i"<br>Ex ia IIC T6              | Pi = 0.2W                                                                  |
|                                                                                | Protection type "Ex ta"                             | U <sub>max</sub> = 25V                                                     |
|                                                                                |                                                     | Gas Zones 0, 1, 2 with safety barriers                                     |
|                                                                                |                                                     | Dust Zones 20, 21, 22 without safety barriers. Maximum supply voltage 25V. |

### Dimensions

| Capacity                   | 15t - 20t | 30t  | 40t  | 50t  | 60t  |
|----------------------------|-----------|------|------|------|------|
| A                          | 200       | 200  | 200  | 200  | 260  |
| B                          | 150       | 150  | 150  | 150  | 210  |
| C                          | 123       | 123  | 123  | 123  | 153  |
| d max **                   | 15.5      | 15.5 | 15.5 | 15.5 | 13.5 |
| e max permitted tilting ** | 6°        | 6°   | 6°   | 6°   | 3.7° |
| r sph. radius              | 130       | 160  | 180  | 180  | 220  |

All dimensions are in mm



### Electrical Connections

Via 6 core, 6mm diameter, screened Polyurethane cable with high toughness and high resistance to rodents

### Construction

Load cell manufactured from stainless steel

### Options

Mounting accessory set comprising top & bottom mounting cups, 3 centering discs, nitrile rubber protective gaiter and securing clip.

Part number

LA34X-50T-NP (15t-50t, nickel plated alloy steel) or LA34X-60T-NP (60t, nickel plated alloy steel)

### Thames Side Sensors Ltd

Unit 10, io Trade Centre, Deacon Way, Reading, Berkshire RG30 6AZ, UK

tel: +44 (0) 118 941 1387

fax: +44 (0) 118 941 2004

sales@thames-side.co.uk

www.thames-side.com

Issue: T34X.11.24



Our policy is one of continuous product enhancement. We therefore reserve the right to incorporate technical modifications without prior notification.



www.thames-side.com